

Emily Cook

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SUMMARY:

Geophysicist experienced in near-surface geophysics, geospatial modeling, energy feasibility, and federal geohazard assessment. Passionate about community-centric challenges across energy, climate, and hazards.

EDUCATION:

Binghamton University, SUNY | May 2026 | B.S. in Geophysics (3.93 GPA) | Summa Cum Laude | Dean's List

SKILLS:

Geophysics: ERT (ResIPy, AGI Earth Imager), SRT (Rayfract), MASW (SurfSeis), GPR (GPRPy), EM, HVSr (Grilla), UAV-TIR (Pix4D, Thermoviewer), Borehole geophysics, LiDAR

Geospatial & Modelling: ArcGIS Pro, QGIS, Leapfrog, Python, Unix, Geodatabase design

Research: Technical Writing, Presentations, Graphics, UTV/ATV, Survey Design, Backcountry Operations

EXPERIENCE:

Geophysics Contractor

August 2026 – Present

United States Geological Survey (USGS)

Coram, NY

- Characterize hydrogeologic systems using geophysical, geospatial, and hydrologic workflows to map groundwater quality, soil dynamics, and aquifer geometry in coastal watersheds.
- Partner with federal and local stakeholders to safeguard public drinking water supplies and support decarbonization and ecological restoration initiatives.

Geophysics & Seismology Intern

May 2026 – August 2026

United States Bureau of Reclamation (BOR)

Denver, CO

- Acquire, process, and invert 2D/3D geophysical data (ERT, SRT, MASW, GPR, tTEM) to characterize canal seepage, spillway voids, and landslide slip planes for federal infrastructure safety.
- Manage and characterize 300+ fault traces, geochronological points, and geophysical surveys in a federal GIS database, directly streamlining engineering risk analysis.

Energy Researcher

May 2025 – Present

Alaska Center of Energy and Power (ACEP)

Fairbanks, AK / Remote

- Create and quantify feasible climate solution scenarios of the Energy and Transportation sector for a comprehensive statewide policy report in collaboration with the Nature Conservancy (TNC).
- Collaborate with non-profits, Alaskan stakeholders, and indigenous communities to ensure projects prioritized energy sovereignty, culminating in a published paper and international presentations.
- Led spatial and multi-criteria decision analysis (MCDA) for five deepwater Alaskan ports, modeling geological and economic feasibility for geothermal e-fuel production for decarbonization.

Direct Use Geothermal Researcher

January – December 2024

Summer Scholars & Artists Program

Binghamton, NY

- Built a centralized regional ArcGIS database, synthesizing depth-to-layer, borehole geophysical logs, and thermal datasets from 1,000+ oil and gas well logs for 3D subsurface characterization to isolate high-probability geothermal potential zones.

Geology Research Assistant

May – August 2024

SUNY RF funded Basin Research Team

Binghamton, NY

- Data acquisition for UAS and water-borne temperature and hydrogeophysical surveys for submarine groundwater discharge (SGD).
- Assisted geophysical field campaigns using SRT and ERT to characterize basin geometry.

Drone Geophysics Researcher

First-Year Immersion Program (FRI)

August 2022 – January 2024

Binghamton, NY

- Investigated SGD's role in sustainable water resource management for island communities using UAV-TIR to assess seasonality of seep, summarizing results in research paper and poster session.
- Conducted remote sensing fieldwork and cartography exercises using various geophysical sensors.

PUBLICATIONS:

Cook, E. & de Witt, M. (2025) Assessing the Feasibility of Geothermal-to-X for Sustainable Maritime Refueling in Alaska. *Clean Technologies*, 7(4), 115. <https://doi.org/10.3390/cleantechnol7040115>

PRESENTATIONS:

Multi-method geophysical analysis of West Canal slope failure	US BOR	2026 Poster
Geothermal-to-X Potential in Alaska's Maritime Industry	Arctic Frontiers	2026 Poster
Geothermal-to-X Potential in Alaska's Maritime Industry	ACEP	2025 Talk
Feasibility of Geothermal-to-X in Alaskan Arctic Shipping	GEODE	2025 Talk
Spatial mapping of geothermal energy variables near Binghamton, NY	SSAP	2024 Talk
Seasonal Variation in Detecting Submarine Groundwater Discharge Using Drone-Mounted Thermal Imaging	FRI Poster Session	2023 Poster
Viability of Coastal Seeps to Combat Droughts on Rapa Nui	FRI Poster Session	2022 Poster

HONORS & AWARDS:

Ralph Digman Award - Outstanding Earth Science Senior	2026
Dean's List (Binghamton)	All
Art of Science Competition "Best in Show" (Binghamton)	2026
Phi Beta Kappa	2025
Best Poster Award (Peer J at Arctic Frontiers)	2026
Harpur Edge High Impact Award (Binghamton)	2026, 2023
Summer Scholars & Artists Program (Binghamton)	2024
SUNY Impact Scholarship (NYS Wireless Association)	2024
Outside the Classroom Award (EF)	2023

PROJECTS:

1. **Guest Science Educator:** Designed and delivered interactive climate and energy presentations connecting the Arctic issues with local issues for K-12 students (2025-2026).
2. **Team Captain (Binghamton Ultimate Frisbee):** Led and managed logistics for ~35 athlete roster for travel and competitive play across East Coast; two National Championship appearances (2024-2026).
3. **Cartographer:** Custom mapping projects (QGIS, ESRI, Leapfrog) and portfolio web pages, including custom LiDAR-art like "Arctic Arteries" in Art of Science competition (2026).
4. **Geophysics Summer School (Texas A&M):** Seismic refraction field methods; 2018 Kilauea waveform analysis and seismicity using MatLab and QGIS (2025).
5. **GEOL 451 and 453:** Independent course projects in applied geophysical methods. Survey acquisition, processing, and interpretation of wood ash on soil over time (ERT, IP, EMI), locating unmarked graves (GPR), and locating abandoned oil and gas wells (smartphone magnetometers) (2024-2025).
6. **Rover Team Geology Lead:** Integration of remote sensing and hydrologic interpretation for autonomous planetary analog missions (2023-2024).
7. **Ecology Field Courses:** Immersive geological and ecological field study in the Everglades and Utah and Wyoming National Parks over five weeks (2023).